

Prior Publication Statement

MEET-COBRA: Machine-learning-based Energy-Efficient proactive Coordination of handOver, Beamforming and Resource Allocation

Bowen Xie, Sheng Zhou, and Zhisheng Niu

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Earlier publication

Bowen Xie, Sheng Zhou, Zhisheng Niu, Hao Wu, and Cong Shi, “AEPHORA: AI/ML-Based Energy-Efficient Proactive Handover and Resource Allocation,” in *ICC 2025 – IEEE International Conference on Communications*, Montreal, Canada, June 8–12, 2025, pp. 6297–6303.

DOI: 10.1109/ICC52391.2025.11161541. The conference was sponsored by the IEEE Communications Society (conference website).

Relationship and main extensions

The journal manuscript extends the above conference paper. Both works consider a two-tier vehicular heterogeneous network and use prediction-assisted proactive handover (PHO) and queue-aware resource allocation (RA) to reduce system transmit power under latency constraints. The main extensions are as follows.

1. **Joint coordination of PHO, beamforming, and RA.** The conference paper uses a scalar channel model with a prescribed beamforming gain and does not optimize beam selection. The journal manuscript introduces microcell multiple-input multiple-output (MIMO) channels, transmit and receive beam-pair selection, and beam-training overhead into the joint optimization. It develops prediction-guided beam search with early termination and coordinates it with frame-level PHO and slot-level RA. It also accounts for inter-cell interference and handover interruption when estimating resource demands and allocating resources.
2. **Prediction of beam pairs and channel gains.** The conference paper predicts vehicle positions from historical channel gains and estimates future link gains from the predicted positions. The journal manuscript instead uses historical superposed-pilot observations to predict next-frame beam-pair candidates, desired-link gains, and beam-averaged interfering-link gains. These predictions guide association decisions, beam search, and resource allocation without requiring full MIMO channel reporting.
3. **Ray-traced channels and broader evaluation.** The conference evaluation combines SUMO vehicle trajectories with statistically generated scalar channels. The journal evaluation uses SUMO mobility traces and Sionna Ray Tracing to generate microcell MIMO channels, and computes receive rates using the selected beams and explicit resource block assignments. It includes comparisons with two adapted joint handover and beamforming methods from the literature, module ablations, multiple random seeds, analyses of mobility and prediction errors, and measurements of neural-network complexity, inference latency, and reporting overhead.

The conference paper is cited as reference [13] in the journal manuscript. Section I, page 2, explicitly identifies the manuscript as an extension and summarizes the principal differences. A copy of the conference paper is provided separately with this statement.